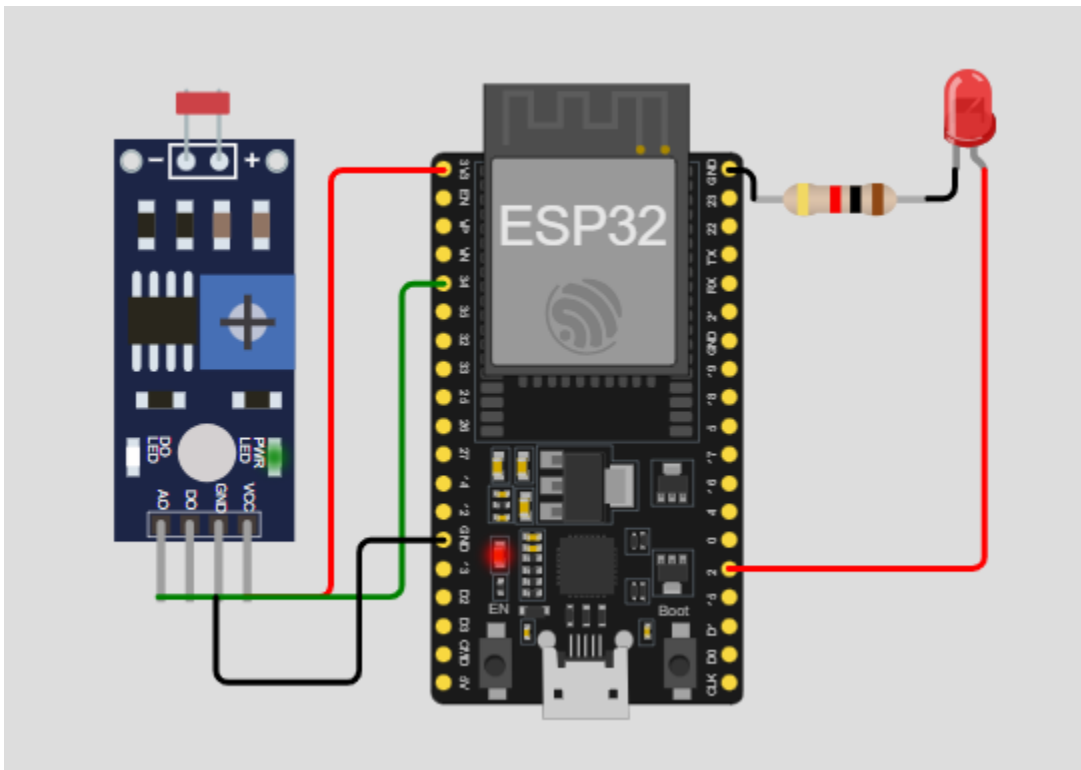


WORKING OVERVIEW

This smart lighting system uses an LDR sensor connected to ESP32 to detect ambient light intensity. The ESP32 reads analog values from the LDR and compares them with a predefined threshold. If the light level is low (dark condition), the ESP32 automatically turns ON an LED, simulating a smart light. If the light level is high (bright condition), the LED is turned OFF. This demonstrates automation using IF-ELSE decision logic. Additionally, the system is integrated with ThingSpeak cloud using WiFi. The ESP32 sends real-time LDR sensor data to the cloud every 15 seconds. This allows remote monitoring of light conditions through graphs and dashboards. The Serial Monitor displays live sensor readings and system status for debugging and feedback. This project represents a basic IoT-enabled smart system widely used in street lighting and home automation applications.

CIRCUIT DIAGRAM :



CONNECTIONS :

LDR SENSOR -

- VCC → 5V
- GND → GND
- AO → A0 (analog pin)

LED -

LED (+) → Digital Pin 9

LED (-) → GND via resistor

SOURCE CODE :

```
#include <WiFi.h>
#include <HTTPClient.h>

// Wokwi default WiFi
const char* ssid = "Wokwi-GUEST";
const char* password = "";

// ThingSpeak
String apiKey = "*****";
const char* server = "http://api.thingspeak.com/update";

// Pins
#define LDR_PIN 34
#define LED_PIN 2

int threshold = 2000; // adjust if needed

void setup() {
    Serial.begin(115200);
    pinMode(LED_PIN, OUTPUT);

    WiFi.begin(ssid, password);
    Serial.print("Connecting");

    while (WiFi.status() != WL_CONNECTED) {
        delay(300);
        Serial.print(".");
    }

    Serial.println("\nWiFi Connected!");
}

void loop() {
    int ldrValue = analogRead(LDR_PIN);

    Serial.print("LDR Value: ");
    Serial.println(ldrValue);
```

```

// Automation Logic
if (ldrValue < threshold) {
    digitalWrite(LED_PIN, HIGH);
    Serial.println("LED ON (Dark)");
} else {
    digitalWrite(LED_PIN, LOW);
    Serial.println("LED OFF (Bright)");
}

if (WiFi.status() == WL_CONNECTED) {
    HTTPClient http;

    String url = String(server) + "?api_key=" + apiKey + "&field1=" + String(ldrValue);

    http.begin(url);
    int httpResponseCode = http.GET();

    Serial.print("HTTP Response: ");
    Serial.println(httpResponseCode);

    http.end();
}

delay(15000); // ThingSpeak limit
}

```

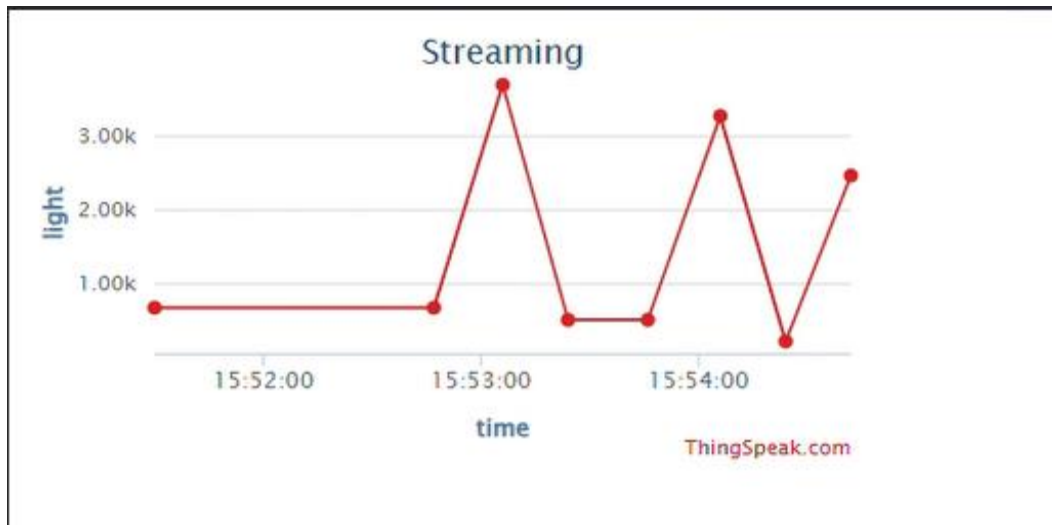
CODE EXPLANATION :

This program implements a smart lighting system using an LDR sensor and an ESP32. The LDR is connected to an analog pin, which continuously reads light intensity values. These values vary depending on the surrounding light conditions—lower values indicate darkness, while higher values indicate brightness. A threshold value is defined in the code to differentiate between dark and bright environments.

Inside the loop function, the ESP32 reads the LDR value using the `analogRead` function and prints it to the Serial Monitor for real-time monitoring. The system then uses an IF-ELSE condition to make decisions. If the LDR value is below the threshold, it means the environment is dark, so the LED is turned ON automatically. If the value is above the threshold, indicating sufficient light, the LED is turned OFF.

Additionally, the ESP32 connects to a WiFi network and sends the LDR data to a cloud platform using an HTTP request. The data is transmitted at regular intervals, allowing remote monitoring of light intensity. This system demonstrates how embedded devices can perform automated decision-making and send real-time data to the cloud, forming a basic Internet of Things (IoT) application.

OUTPUT RESULT ON CLOUD :



OVERVIEW

This project presents a Smart Lighting System using an LDR sensor and ESP32 with basic IoT functionality. The system automatically controls lighting based on ambient light conditions. The LDR sensor detects light intensity and sends analog values to the ESP32. A threshold value is set in the program to distinguish between dark and bright environments. When the light level falls below the threshold, the ESP32 turns ON an LED, simulating a smart light. When the light level is high, the LED turns OFF, ensuring energy efficiency.

In addition to automation, the system includes cloud integration for remote monitoring. The ESP32 connects to a WiFi network and sends real-time LDR data to a cloud platform like ThingSpeak using HTTP requests. The data is displayed in graphical form, allowing users to monitor light intensity from anywhere. This project demonstrates embedded system decision-making using IF-ELSE logic and introduces basic IoT concepts such as data streaming, remote access, and real-time monitoring, making it suitable for applications like smart homes and automatic street lighting systems.